10/8/26

6. No. Consider the "central" cube (the one which, if the large cube were painted before cutting, would be the only one without paint after cutting)
In order to free this central cube, all 6 of it's faces need to be cut at some point. Since there is no way for a single planar cut to cut 2+ of it's faces at once, the minimum number of cuts to free this cube is 6, and therefore the minimum for the whole task is not less than 6.